

Learning objectives

In this lesson you will learn to:

- Describe how devices are connected on a wireless network
- Compare the performance of wired and wireless LANs and give examples of situations where one is preferable to the other
- Summarise the characteristics of Wi-Fi, Bluetooth[®], RFiD, Zigbee and NFC and give examples of their use.

For more information and additional learner activities see Chapter 4.1 of the student book.

Wireless networks

Last lesson you learned about how wired connections allow the transmission of data.

You also learned about the differences between wired and wireless connections.

In this lesson you will learn about the different types of wireless transmission.

A bit of science

In science you will have learned about electromagnetic waves.

Microwaves and radio waves are frequently used to transmit data. This involves converting the binary data into an electromagnetic wave using a specific protocol.

Wi-Fi and Bluetooth® are examples of the protocols that govern the conversion of data.

Limitations

Wireless data transmission has its limits.

- Generally the range of wireless transmission is limited to a few metres.
- It is easily interrupted by other signals or physical obstacles.
- It has much lower bandwidth than wired connections which makes transfer slow.
- Security is a concern as wireless signals can be intercepted by others.

WLAN

Previously you learned that a LAN is a network limited to a single building or site.

WLANs are **Wireless Local Area Networks**.

You probably have a WLAN at home.

If you use Wi-Fi to connect your devices, that's a WLAN.

Wireless access point

In a WLAN, there will be a **wireless access point (WAP)** which sends and receives wireless transmissions to and from the other devices.

A WAP is usually part of a box you call the **broadband router**.

The router will be connected (usually wired) to an internet access point in order to connect the WLAN to the internet.

The best option

Most LANs are generally a combination of wired and wireless technologies.



Other technologies

Wi-Fi is not the only common wireless data transmission technology. You will probably have experienced other connection types including:

- Bluetooth®
- Radio Frequency Identification (RFID)
- Near Field Communication (NFC)

And less commonly, if you have a digital personal assistant device such as Alexa, a communication technology called Zigbee.

Wrap up: you have learned how to...

- ✓ Describe how devices are connected on a wireless network.
 - Wi-Fi uses radio waves to transmit data to and from a wireless access point.
 - The router, connected to the wireless access point, distributes signals to the devices (unless any of them uses Wi-Fi Direct, which does not need the router).
- ✓ Compare the performance of wired and wireless LANs and give examples of situations where one is preferable to the other.
 - Range, interference, ease and cost of installation, latency, flexibility, reliability, and bandwidth all need to be considered.
- ✓ Summarise the characteristics of Wi-Fi, Bluetooth®, RFiD, Zigbee and NFC and give examples of their use.